

SOFTWARE CHANGE REQUEST

CR-019: Student Profile Editing

1. Change Request Details

Change Request ID	CR-019
Title	Student Profile Editing
Priority	Medium
Difficulty	Medium
Change Type	Functional Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing GET /api/students/me endpoint, authentication middleware, and existing student profile data

2. Business Requirement

Currently, the Student Management System allows an authenticated student to view their own profile through the Student Dashboard, but there is no student-facing facility to edit the displayed profile information. The system should allow a student to update their own email address and phone number without administrator assistance.

3. Current Behaviour

- StudentDashboardPage.jsx loads the authenticated student's profile using GET /api/students/me.
- The My Profile section displays registration number, full name, email, phone number and member-since date.
- The displayed profile information is read-only; no Edit Profile button or student update form is available.
- The existing PUT /api/students/:id update route is protected for ADMIN users and is intended for administrator updates.
- Student authentication is handled by the existing JWT authentication middleware.

4. Requested Change

Add a student-facing Edit Profile facility to the My Profile section. The implementation must:

- Add an Edit Profile button to StudentDashboardPage.jsx.
- Provide a form for editing the student's email address and phone number.
- Pre-populate the form with the student's current values.
- Validate email and phone number before submission.
- Add a student-authorized profile update API, preferably PUT /api/students/me, rather than exposing a student_id for self-update.
- Identify the student from the authenticated JWT userId.

- Update only the authenticated student's existing record.
- Return a clear success or error response.
- Refresh the displayed profile information after a successful update.
- Preserve the existing GET /api/students/me profile retrieval behaviour.
- Preserve existing administrator CRUD and authorization behaviour.

5. Scope

In Scope

- Edit Profile button and student edit form.
- Email address update.
- Phone number update.
- Client-side and server-side validation.
- Authenticated student profile update endpoint.
- Database update of existing email and phone_number fields.
- Success and error messages.
- Refresh/display of updated profile information.
- Responsive profile editing interface.

Out of Scope

- Changing registration number.
- Changing student password.
- Changing authentication or login mechanism.
- Changing administrator CRUD functionality.
- Adding a new student record.
- Database schema redesign.
- Updating another student's profile.

6. Expected / Potentially Affected Files

- frontend/src/pages/StudentDashboardPage.jsx
- frontend/src/services/api.js (existing API utility; likely reused without structural change)
- frontend/src/styles/studentDashboard.css
- backend/src/routes/studentRoutes.js
- backend/src/controllers/studentController.js
- backend/src/models/studentModel.js

Any additional files modified during implementation must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

Functional Requirements

- FR-001: An Edit Profile option shall be available on the student's My Profile page.
- FR-002: The form shall display the current email address and phone number.
- FR-003: Students shall be able to update their email address with valid input.
- FR-004: Students shall be able to update their phone number with valid input.
- FR-005: Invalid input shall be rejected with a clear validation message.
- FR-006: Valid changes shall be stored in the existing student record.
- FR-007: The profile view shall display the updated values after a successful update.
- FR-008: Existing GET /api/students/me behaviour shall continue to work.

Security Requirements

- SEC-001: Existing JWT authentication shall remain in use.
- SEC-002: Only an authenticated student shall be able to update a profile.
- SEC-003: The update operation shall derive the student identity from the authenticated user context.
- SEC-004: A student shall not be able to update another student's record by supplying a different student_id.
- SEC-005: Existing administrator authorization and CRUD controls shall remain unchanged.
- SEC-006: Input shall be validated before database update.

8. Acceptance Criteria

- AC-001: The student can open the Edit Profile form from My Profile.
- AC-002: Current email and phone values are pre-filled.
- AC-003: A valid email address can be saved successfully.
- AC-004: A valid phone number can be saved successfully.
- AC-005: Invalid email/phone input is rejected with a clear message.
- AC-006: Updated values persist in the database.
- AC-007: The updated values are displayed after saving.
- AC-008: A student cannot update another student's profile.
- AC-009: Existing student login and profile retrieval continue to work.
- AC-010: Existing administrator CRUD functionality continues to work.

9. Expected Outcome

Students can securely maintain their own email address and phone number through the Student Management System. The change improves usability while retaining the existing authentication, authorization, administrator CRUD and database structure.

10. Implementation Report Requirement

- Implementation summary.

- Affected files and justification.
- Frontend, backend/API and database changes.
- Technical and security changes.
- Test cases and results.
- Acceptance criteria verification.
- Relevant screenshots/evidence.
- Final status and outstanding issues.